

Homework 7

Solve the following problems justifying all your answers.

1. Solve the following integrals

- (a) $\int_0^{\frac{\pi}{2}} \cos^2(x) dx$
- (b) $\int \sin^3(\theta) \cos^4(\theta) d\theta$
- (c) $\int_0^{\frac{\pi}{4}} \tan^4(\theta) d\theta,$
- (d) $\int e^{\sin(t)} \cos(t) dt$
- (e) $\int t \cos^5(t^2) dt$
- (f) $\int \tan^2(\theta) \sec^4(\theta) d\theta$
- (g) $\int_0^{\frac{\pi}{2}} \frac{\cos(t)}{\sqrt{1 + \sin^2(t)}} dt$
- (h) $\int t \sin^2(t) dt$

2. Solve the following integrals

- (a) $\int \frac{x^2}{\sqrt{9 - x^2}} dx$
- (b) $\int \frac{x^2}{(3 + 4x - 4x^2)^{\frac{3}{2}}} dx$
- (c) $\int \frac{3t - 2}{t + 1} dt$
- (d) $\int_1^2 \frac{x^3 + 4x^2 + x - 1}{x^3 + x^2} dx$
- (e) $\int_0^{0.6} \frac{x^2}{\sqrt{9 - 25x^2}} dx$
- (f) $\int \frac{1}{(t^3 - 1)^2} dt$
- (g) $\int \frac{e^{2x}}{e^{2x} + 3e^x + 2} dx$
- (h) $\int \frac{x^4 + 8x^2 + x + 16}{x(x^2 + 4)^2} dx$

3. Use a substitution and then integration by parts to solve

$$\int \frac{\sin(\ln(\sqrt{1 + x^{-4/5}}))}{(1 + x^{-4/5})(x + x^{9/5})} dx$$

4. Show that if we rotate the curve $y = e^{\frac{x}{2}} + e^{-\frac{x}{2}}$ about the x -axis, the area of the resulting surface is the same value as the enclosed volume for any interval $[a, b]$.
5. For the function $f(x) = \frac{1}{4}e^x + e^{-x}$, prove that the arc length on any interval has the same value as the under the curve.
6. Solve the following integrals:

(a) $\int_0^\infty \sin x \, e^{\cos x} \, dx$

(b) $\int_0^2 \frac{dt}{\sqrt{4+t^2}}$

(c) $\int_0^1 \frac{e^{\frac{1}{x}}}{x^3} dx$